Final – MATH 346/377 – Winter 2025

DATE: Tuesday, April 15 TIME: 9:00AM – 12:00PM

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ASSOCIATE EXAMINER: Henri DARMON

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

- There are 11 problems, each worth 10 points. For Math 346, only your 8 best problems will count, for a total of 80 points. For Math 377, your 9 best problems will count, for a total of 90 points.
- Notes, calculators and other electronic devices are not permitted.

This page will not be graded and can be used as scrap paper.

1. (10 points) True or false? Justify your answer.

   (a) Let p be an odd prime. If g is a primitive root modulo p, then it is not a quadratic residue modulo p.

   (b) $x = -\frac{1}{3}$ is a 3-adic *integer* (in $\mathbb{Z}_3$).

   (c) Let p be a prime and a an integer. If $\gcd(a, p^2) = p$ then $\gcd(a + p, p^2) = p$.

   (d) There exists an integer n for which the system $7x + (4n)y = 13$ has no solutions $x, y \in \mathbb{Z}$.

2. (10 points) Show that the equation

$$g(x) = (x^2 - 5)(x^2 - 41)(x^2 - 205)$$

has no integer solutions, but the congruence $g(x) \equiv 0 \mod p$ has a solution for every prime p.

3. (10 points) Find all integers $0 \leq x \leq 24$ that satisfy

$$x^3 \equiv 13 \mod 25.$$

4. (a) (6 points) Find all residues of order 5 modulo 31.

   (b) (4 points) Deduce all integers $1 \leq x \leq 30$ solving

$$1 + x + x^2 + x^3 + x^4 \equiv 0 \mod 31.$$

5. (10 points) What are the last two digits of $(2357)^{3^{20}}$? Justify your answer.

6. (10 points) Find all integers $0 \leq x \leq 92$ that solve the equation

$$31x^{30} + 5x - 3 \equiv 0 \mod 93 = 3 \cdot 31.$$

7. (a) (5 points) For which primes p is $p^2 + 1$ prime?

   (b) (5 points) For which primes p is $p^2 + 2$ prime?

8. (10 points) Use Euclid's idea to prove (by contradiction) that there are infinitely many primes congruent to 3 modulo 4.

9. (a) (5 points) Let g be a primitive root modulo a prime $p \equiv 1 \mod 4$. Show that $g^{\frac{p-1}{4}}$ is a root of $x^2 + 1 \equiv 0 \mod p$.

   (b) (5 points) Use part (a) to find the two roots of $x^2 + 1 \equiv 0 \mod 13$.

10. (a) (8 points) Show that the Euler ϕ-function satisfies $\phi(n^2) = n\phi(n)$ for any integer $n \geq 1$. Deduce that for $k \geq 2$ and p a prime

$$\phi(\phi(p^k)) = p^{k-2}\phi((p-1)^2).$$

(b) (2 points) Compute $\phi(\phi(5^3))$.

11. For an odd prime p we consider the sum

$$S(p) := \sum_{k=1}^{p-1} \left(\frac{k}{p}\right).$$

(a) (4 points) For $p = 5$, verify that $S(5) = 0$.

(b) (6 points) Show that $S(p) = 0$ for all odd primes.